



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Subject: Programming with Java

Sem: III **AY:** 2026-27

Assignment No: 14

TITLE : Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Problem Statement:

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

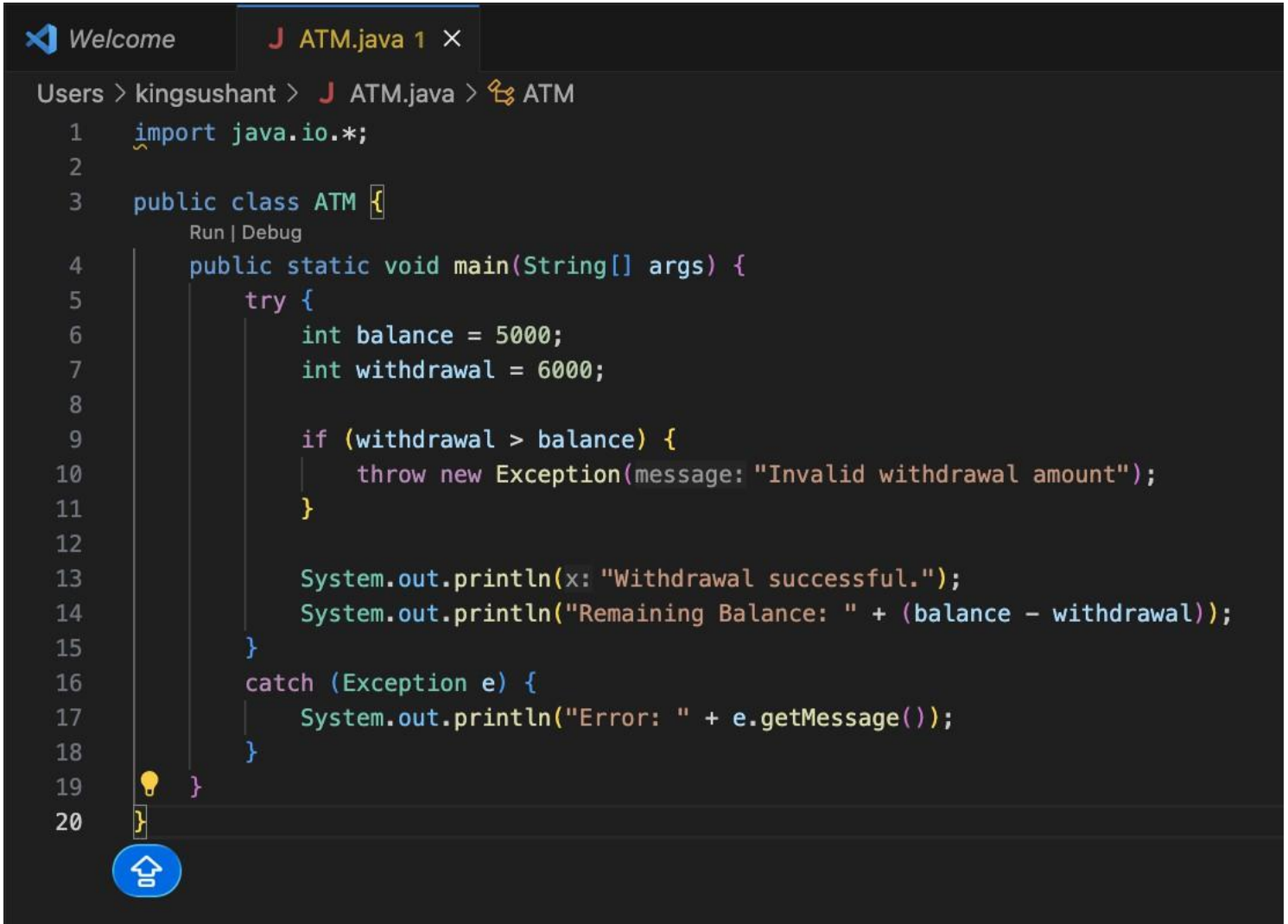
To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory

Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

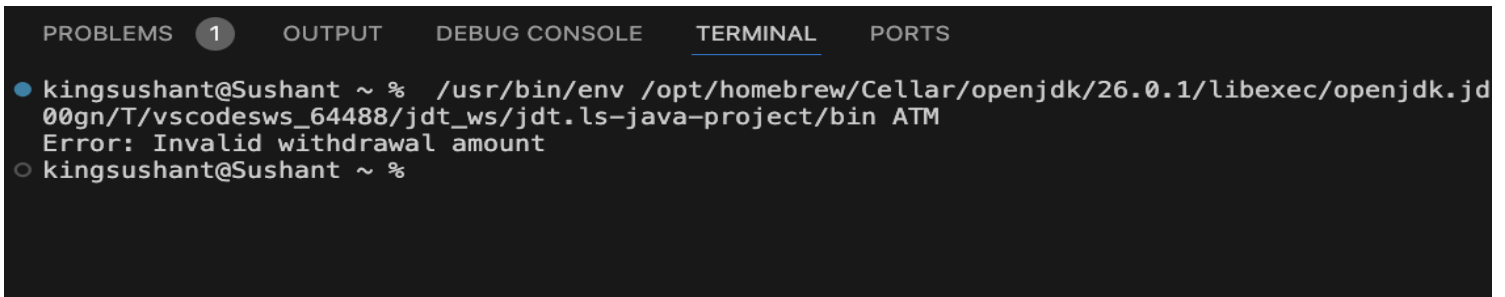
Exercises:

1. Create an ATM program handling invalid withdrawal amount using try-catch.



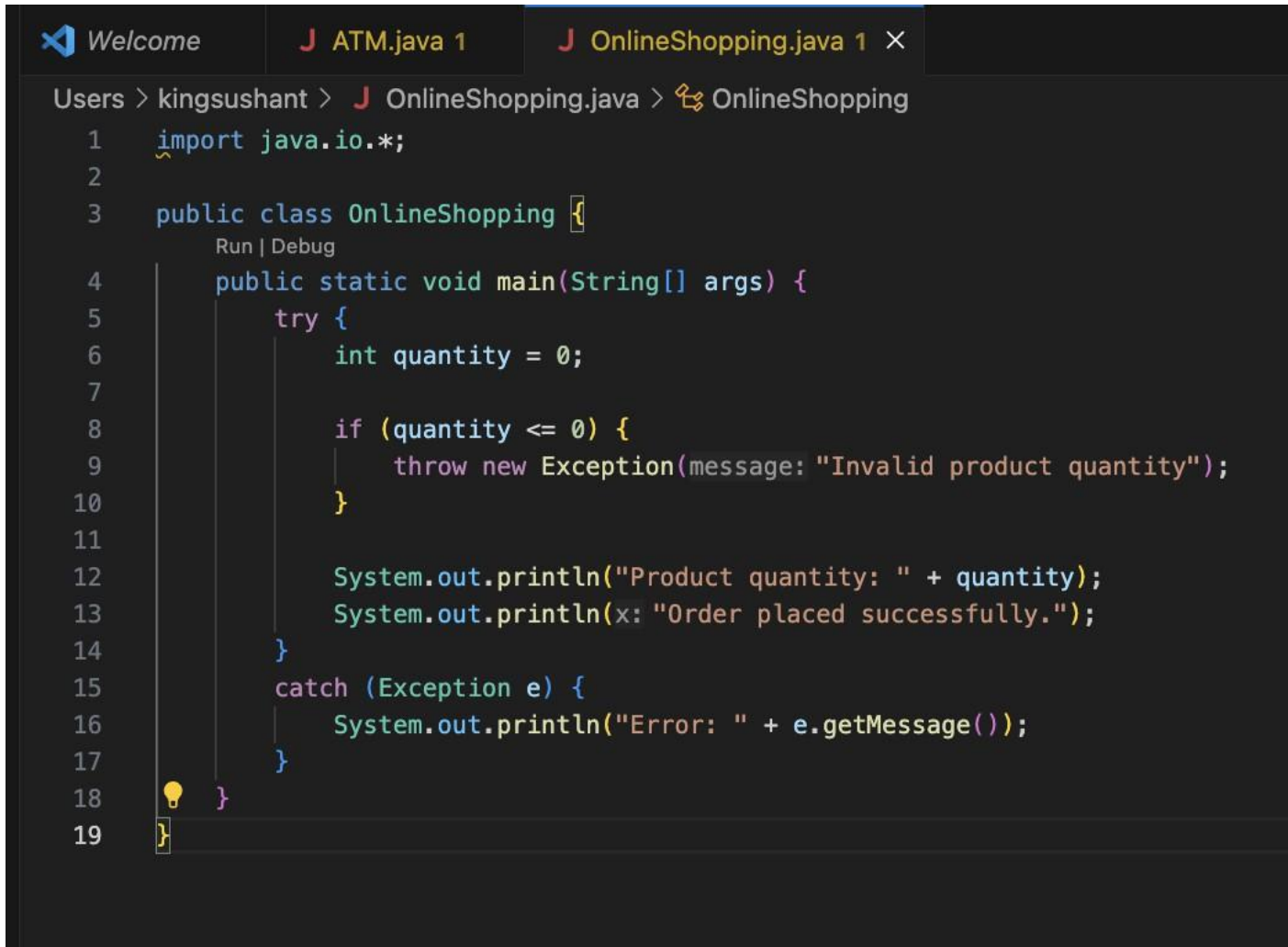
```
1  import java.io.*;
2
3  public class ATM {
4      public static void main(String[] args) {
5          try {
6              int balance = 5000;
7              int withdrawal = 6000;
8
9              if (withdrawal > balance) {
10                 throw new Exception(message: "Invalid withdrawal amount");
11             }
12
13             System.out.println("Withdrawal successful.");
14             System.out.println("Remaining Balance: " + (balance - withdrawal));
15         }
16         catch (Exception e) {
17             System.out.println("Error: " + e.getMessage());
18         }
19     }
20 }
```

Output:



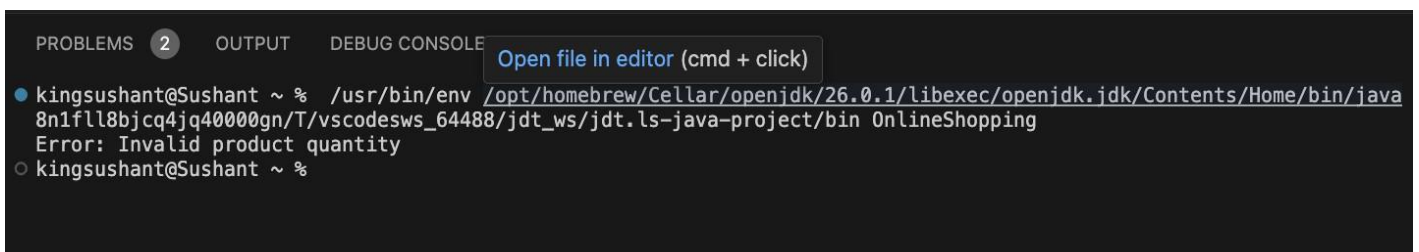
```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
● kingsushant@Sushant ~ % /usr/bin/env /opt/homebrew/Cellar/openjdk/26.0.1/libexec/openjdk.jdk
00gn/T/vscodesws_64488/jdt_ws/jdt.ls-java-project/bin ATM
Error: Invalid withdrawal amount
○ kingsushant@Sushant ~ %
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.



```
1  import java.io.*;
2
3  public class OnlineShopping {
4      public static void main(String[] args) {
5          try {
6              int quantity = 0;
7
8              if (quantity <= 0) {
9                  throw new Exception(message: "Invalid product quantity");
10             }
11
12             System.out.println("Product quantity: " + quantity);
13             System.out.println(x: "Order placed successfully.");
14         }
15         catch (Exception e) {
16             System.out.println("Error: " + e.getMessage());
17         }
18     }
19 }
```

Output:



```
PROBLEMS 2 OUTPUT DEBUG CONSOLE
Open file in editor (cmd + click)
kingsushant@Sushant ~ % /usr/bin/env /opt/homebrew/Cellar/openjdk/26.0.1/libexec/openjdk.jdk/Contents/Home/bin/java
8n1fl18bjcq4jq40000gn/T/vscodews_64488/jdt_ws/jdt.ls-java-project/bin OnlineShopping
Error: Invalid product quantity
kingsushant@Sushant ~ %
```

Conclusion:

The programs successfully demonstrate exception handling using try and catch blocks in Java. The ATM program handles an invalid withdrawal amount, while the Online Shopping program handles a product quantity less than or equal to zero. The catch block displays appropriate error messages when exceptions occur. These programs show how exception handling prevents abnormal termination and makes applications more reliable.